# ECO-PULSE
## Final Evidence Sheet

**AESH Sustainability Hackathon 2026** · Challenge 2: Green Radar Systems · Egypt, July 2026

### Problem Statement

Flash floods in Egyptian wadis develop with little warning; fixed-power, always-on radar monitoring wastes energy at unattended, solar-powered sites and is not sustainable for long-term deployment.

### Core Function

A **24 GHz ground radar** tracks wadi water level and rate-of-rise in real time (matched filter → CFAR → Kalman tracking → ARIMA/LSTM forecasting → threshold+ML decision logic), and switches between a low-power Zadoff-Chu "calm" waveform (Mode 1) and a higher-power LFM chirp "track" waveform (Mode 2) based on assessed risk.

### Baseline Case

**Fixed-duty-cycle operation:**
The radar always transmits at the same rate/power regardless of risk level (`energy_adaptive_enabled = false`).

### Improved Case (Green)

**Adaptive duty-cycling** (`energy_adaptive_enabled = true`): 60% duty while calm, dropping to 30% under low battery (<20%), rising to 100% duty / Mode 2 the instant risk crosses threshold. Validated across all 5 scenarios (`stable`, `slow_rise`, `fast_rise`, `receding`, `flash_flood`) with fixed random seeds for a fair comparison.

### Primary Technical KPI

- **Two-Stage adaptive CFAR:** false-alarm cells per run reduced 5.05 → 4.37 (**-13.5%**) at unchanged 100% detection rate (200-trial stress test).[cite: 1]
- **Forecast accuracy:** 30 s-horizon RMSE improved from 0.761 m (ARIMA baseline) to 0.370 m with the optional LSTM forecaster (**-51%**).

### Primary Sustainability KPI

Adaptive duty-cycling **reduces transmit energy** during calm periods (60% duty, or 30% under low battery) versus fixed 100% duty in the conventional baseline, while returning to full 100%/Mode-2 duty immediately when risk is elevated. Exact aggregate % saved per scenario is generated live by `compare_baseline_vs_green.m` / `run_full_validation.m` for the demo run.

### Test Conditions

160 s simulated runs, 24 GHz carrier, AWGN + wave clutter + bank multipath, Pierson-Moskowitz sea-surface model, 30 W solar panel / 50 Wh battery. Reproducible via `compare_baseline_vs_green.m` and `run_full_validation.m`.

### Project Limitations (Disclosed)

At the default fs = 10 kHz, the raw Mode-2 chirp waveform is under-sampled (documented root cause, measured Peak Sidelobe Ratio -0.45 dB vs a healthy -13 dB) and default multipath makes naive range peak-picking ambiguous. Neither affects the validated decision-layer KPIs above, which run on Kalman-filtered/derived quantities — full detail and a tested, opt-in `signal_based` alternative are in `KNOWN_LIMITATIONS.md`.

**Repository Link:** [ECO-PULSE-REPO]